Predicting vortex shedding behind a circular cylinder with physics-informed neural networks

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Abstract

A data-free physics-informed neural network (PINN) applied to the unsteady wake of a circular cylinder is known to collapse onto the steady symmetric solution: the steady field satisfies the governing equations exactly, and the training loss has no notion of its physical instability. This project reproduces that collapse on purpose-generated, validated ground truth, and then determines which of three candidate mechanisms defeats it. On a lattice Boltzmann dataset at Re = 100 (St within 3.3% of published benchmarks, with a C_d overshoot of 4.8% traced to domain confinement), a plain PINN retains 10.6% of the reference wake amplitude. Adding Fourier features or a Reynolds decomposition does not help. *Causal time-marching alone restores shedding data-free*, at 89.2% retained amplitude, and an equal-budget control rules out extra optimiser steps as the cause. The result holds at every Reynolds number tested from 60 to 180: the control keeps 10–43% of the wake, the mechanism-equipped network 88–104%. The best data-free configuration, marched over five shedding periods, predicts the shedding frequency to within 0.1%, at 94.8% of the correct amplitude and 2.4° of phase, with a velocity-field error of 3.6%. A second contribution adds a pressure-free physics residual to a Fourier Neural Operator trained after the method of Renn et al.; the physics term lowers forecast error by 21–28% out to one shedding period, after which both variants have failed.

1 Introduction and objective

Above Re $\approx$ 47 the wake behind a circular cylinder becomes unstable and sheds a Kármán vortex street at a frequency characterised by the Strouhal number St $= fD/U$. Predicting that shedding with a neural network is not merely a question of accuracy, because of a structural difficulty:

1. **The steady solution remains a valid solution.** A symmetric, non-shedding field satisfies the Navier–Stokes equations and every boundary condition exactly. It is unstable physically, but a PINN loss measures residuals, not stability, so the steady field is an easily reachable minimum.
2. **Spectral bias.** Networks fit low-frequency content long before high-frequency content, which directly opposes representing an oscillatory wake.
3. **The initial condition is a measure-zero slice** of the space–time domain, and is therefore weakly weighted against the bulk PDE residual that fills it.

The documented consequence [1] is that a data-free PINN "behaves like a steady solver", while a data-driven one sheds only where data is present. Two points of scope. Their cylinder case is at Re = 200, solved as a *single* time window spanning forty shedding periods, whereas this project reproduces the phenomenon at Re = 100 — and across Re = 60–180 in Section 7

— from a baseline that solves one window of one period and still loses most of the wake, which is a considerably more favourable configuration than theirs. And they attribute the collapse to spectral bias, on the evidence of a Koopman analysis showing damped modes; the ablation below finds that attacking spectral bias directly does not, on its own, keep more of the wake, while enforcing causality does — and does most in combination with the other two mechanisms.

The objective is therefore to reproduce that failure on validated ground truth of our own, and then to establish *which* mechanism defeats it, with an ablation controlled well enough that the answer is attributable. The mechanism that works, causal time-marching, is not new in itself — it is the sequence-to-sequence formulation of Krishnapriyan et al. [4], demonstrated there on one-dimensional convection, reaction and reaction–diffusion problems, and a hard-window relative of the causal weighting of Wang et al. [5]. What is measured here is how much it contributes on two-dimensional wake shedding without data, separated from the extra training a marched run receives, and how it interacts with the other two mechanisms. Both Chuang and Barba [1] and, more recently, Shah [3] — who ablates six techniques on a confined cylinder benchmark and finds that only supervised combinations escape the collapse — report that some form of data supervision is necessary to sustain shedding at all. Neither tests hard time-window marching. The answer found here sits between: a data-free marched chain does sustain a wake, at three-quarters of its amplitude, but only with all three mechanisms together, and over five periods it drifts about half a cycle out of phase with the reference.

A second thread connects the work to its reference paper. Renn et al. [2] forecast subcritical cylinder wakes with a Fourier Neural Operator (FNO) that is *purely data-driven* — there is no physics term anywhere in it. Whether adding physics to such an operator improves forecast stability is the natural question joining the two topics, and is addressed in Section 8.

Reynolds number. The ablation and the accuracy study are at $\mathrm{Re} = 100$: laminar and unambiguously two-dimensional. Renn et al.'s $\mathrm{Re} = 240$–3060 is genuinely three-dimensional, where a 2-D model would be solving the wrong physics. Section 7 repeats the central comparison across $\mathrm{Re} = 60$–180, the upper limit set by the three-dimensional transition at $\mathrm{Re}_c = 188.5 \pm 1.0$ [13].

Software. The work builds on `underPINN`, a JAX PINN framework, which supplies the network architectures, the geometry primitives and a steady Navier–Stokes residual. The unsteady operator (Section 3), the lattice Boltzmann solver, the ablation machinery and the operator-forecasting code are contributions of this project.

2 Ground truth by lattice Boltzmann

No experimental data was available, so the reference wake is generated. The solver (`src/cfd/lbm_cylinder.py`) is a D2Q9 two-relaxation-time (TRT) lattice Boltzmann method written in JAX.

Why these particular choices.

- **TRT collision.** A single-relaxation-time (BGK) scheme places the bounce-back wall at a viscosity-dependent position, so the cylinder's effective diameter drifts with Re. TRT relaxes the symmetric and antisymmetric parts separately, allowing the magic parameter $\Lambda = (\tau^+ - 1/2)(\tau^- - 1/2)$ to be pinned at $3/16$, which fixes the wall location.
- **Half-way bounce-back** for the no-slip cylinder, placing the wall midway between nodes, and **momentum exchange** for the surface force.
- **Specular (free-slip) top and bottom walls**, enforcing $v = 0$ exactly and matching the symmetry condition used on the PINN side.

- **An absorbing layer** of $8D$ at each free-slip wall, relaxing the populations toward the free-stream equilibrium at a peak rate of 0.05 per step on a cubic ramp, with τ^- recomputed pointwise so that Λ stays exactly 3/16. Why it is needed is the subject of the rest of this section.
- **The inlet $25D$ ahead of the cylinder**, in a domain $60D$ long.

Shedding is triggered by a brief transverse inlet perturbation; without it the symmetric base flow persists indefinitely — the same degeneracy that the PINN later falls into, appearing here in the reference solver.

Production dataset. $\mathrm{Re} = 100$, $D = 24$ lattice units, $\mathrm{Ma} = 0.087$, grid 1440×1200, blockage $D/H = 2\%$. The saved record is 641 snapshots covering 20 shedding periods, on a 241×121 grid at $\Delta x = 0.083D$ and $\Delta t^* = 0.19$, chosen near the reference paper's PIV resolution so that the later comparison is like-for-like.

Table 1: Validation of the ground truth at $\mathrm{Re} = 100$. Benchmarks are Qu et al. [11]: the range is their compilation of eight independent large-domain studies, and the errors are taken against their own values, $C_d = 1.317$, $\mathrm{St} = 0.1649$ and $C_l' = 0.2224$.

quantity	computed	benchmark [11]	error
C_d (mean drag coefficient)	1.3634	1.318–1.336	+3.5%
St (Strouhal number)	0.1648	0.164–0.166	−0.04%
C_l' (r.m.s. lift)	0.2362	0.222–0.235	+6.2%

St is measured from wake vorticity, which cannot see irrotational sound; the lift and a single wake probe give the same value to within 0.002%. The remaining C_d offset is resolution, quantified below: at $D = 32$ the same geometry gives +2.3%.

2.1 How the ground truth was validated

An earlier version of this dataset put the inlet $8D$ ahead of the cylinder and had no absorbing layer. It read $\mathrm{St} = 0.1595$ (-3.3%) and $C_d = 1.3798$, and was reported as agreeing with a benchmark of 1.37–1.38 to 0.3%; that benchmark was uncited and closer to the $\mathrm{Re} = 200$ value. Three problems were found and fixed, each by a controlled test (Figure 1).

The domain is a duct, and the duct resonates. Free-slip walls reflect sound perfectly and lattice Boltzmann carries real sound, so the domain supports a transverse standing half-wave at $c_s/2H$ — in flow units $\mathrm{St}_a = (c_s/2H)(D/U) = 0.1155$. Given time it captures the wake itself: with no absorbing layer, wake vorticity sheds at $\mathrm{St} = 0.1600$ over 20 recorded periods and at 0.1132 over 38, on the resonance. A viscosity ramp, the textbook absorbing layer, does nothing here (0.1131): viscous absorption scales as $\nu\omega^2$ and this mode's wavelength is twice the domain height, so an $8D$ layer at twenty times the viscosity removes about 0.04% of the amplitude per pass. Relaxing toward the free stream damps at a rate set per step, independent of frequency, and releases the wake (0.1694 over the same 38 periods). The result does not depend on the layer: across a twofold change in its width and a fivefold change in its rate, C_d moves by 0.1% and St by 0.2%, and 38 and 120 recorded periods agree to four figures.

This is a known physical mechanism rather than a numerical curiosity. Blevins [15] showed experimentally that a transverse acoustic wave shifts cylinder shedding to its own frequency through the acoustic velocity rather than the pressure, and Hong et al. [14] model the coupling for a cylinder in a duct, placing the lock-in band at $\mathrm{St}/\mathrm{St}_a \leq 1.14$. At $\mathrm{Re} = 100$ the ratio here is 1.43, well outside that band, and the wake still locks given enough time — plausibly

because lossless walls sustain the mode at separations a physical duct could not. The practical consequence was that no dataset could be recorded for long enough to average force coefficients over the 80-odd cycles the literature uses.

The inlet distance, not the blockage. With the absorbing layer in place and the grid converged, crossing domain height ($H = 50D$, $120D$) with inlet distance ($8D$, $25D$) separates the two geometric suspects cleanly. Moving the inlet from $8D$ to $25D$ ahead of the cylinder lowers C_d by 4.7%, from +7.4% to +2.3% against Qu et al., and takes St from +3.0% to +0.3%; raising the height from $50D$ to $120D$ moves C_d by 0.36%, and by 0.07% once the inlet is far enough upstream. Posdziech and Grundmann, as reported by Qu et al. [11], require more than $20D$. Blockage, which an earlier draft of this report blamed, was never the dominant error.

Resolution. At fixed geometry, $C_d = 1.4334$, 1.4139 and 1.4147 at $D = 24$, 32 and 48: converged at $D = 32$, with $D = 24$ about 1.4% high in C_d and within 0.3% in St. The production dataset keeps $D = 24$ so that its snapshot grid matches the PIV-like resolution the networks are trained on; the price is the C_d offset in Table 1.

Residual divergence. Lattice Boltzmann is weakly compressible, so $\nabla \cdot \mathbf{u}$ is never zero. Over the snapshot grid, excluding the cylinder and a two-cell halo, its r.m.s. is 2.46% of $|\nabla \mathbf{u}|$, against $\mathrm{Ma}^2 = 0.75\%$. A spectral decomposition separates two parts. The time-mean part, the steady compressibility of the mean flow, is unchanged from the earlier dataset. The fluctuating part is broadband and strongest in the far wake; it is a property of the longer domain, not of the absorbing layer — the same geometry without the layer carries the same level — and the coherent duct mode that dominated it before is gone.

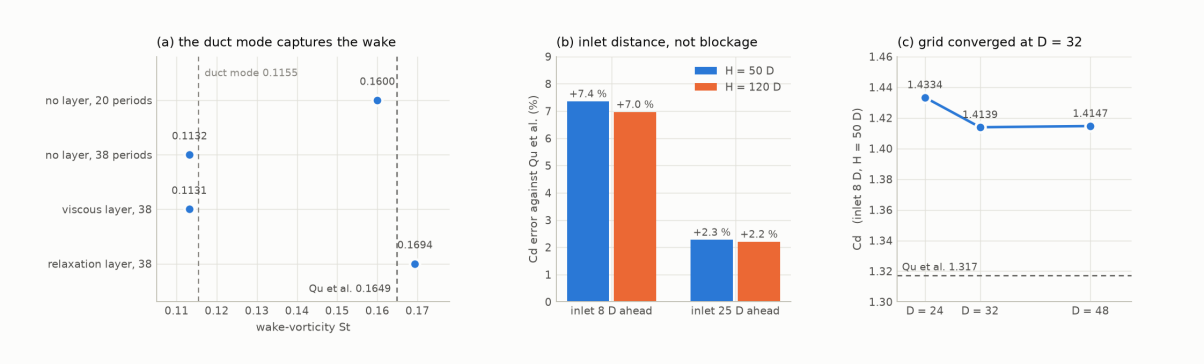


Figure 1: How the ground truth was validated. (a) Wake-vorticity St: without an absorbing layer the wake locks onto the transverse duct mode given enough time; a viscous layer does not prevent it; relaxation toward the free stream does. (b) C_d error at $D = 32$ with the layer: the inlet distance, not the domain height, was the dominant geometric error. (c) Grid convergence at the original inlet distance.

A methodological note. An early estimate of St taken from an FFT peak gave a fraction-of-a-percent agreement that was pure luck: over a record of this length a single FFT bin is wider than the discrepancy being measured. The St values reported here are measured from wake vorticity averaged over many points, cross-checked against a single wake probe and the lift; on the production dataset the three agree to 0.002%, where on the earlier one they disagreed by about 1% because the lift carried the duct mode.

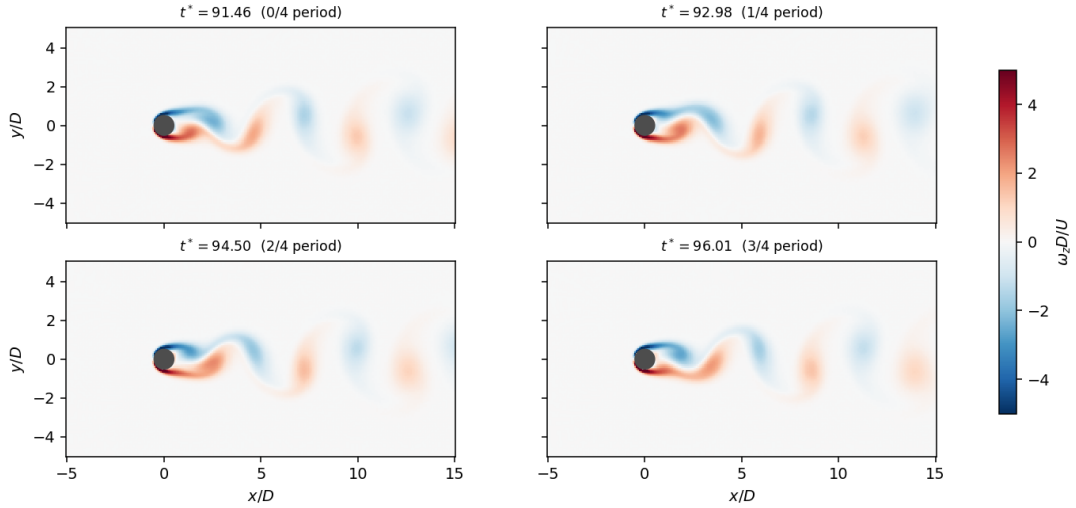


Figure 2: The reference Kármán street at quarter-period intervals, the start of the record. These four instants are the four start phases of Section 4.

3 An unsteady Navier–Stokes operator

`underPINN` ships only a steady incompressible residual, which cannot represent shedding at all. This project supplies the unsteady operator (`src/pinn/ns2d_unsteady.py`), conforming to the framework's `BasePDE` contract so it composes with the existing solvers, samplers and networks. Non-dimensionalised on D and U,

$$u_x + v_y = 0, \tag{1}$$

$$u_t + uu_x + vu_y + p_x - \tfrac{1}{\mathrm{Re}}(u_{xx} + u_{yy}) = 0, \tag{2}$$

$$v_t + uv_x + vv_y + p_y - \tfrac{1}{\mathrm{Re}}(v_{xx} + v_{yy}) = 0, \tag{3}$$

with the network mapping $(x, y, t) \mapsto (u, v, p)$.

Two derivative backends are implemented: a literal `jax.hessian` reference, and a forward-over-forward JVP path computing only the six derivative groups the residual needs, where a full Hessian would compute 27.

Table 2: Verification of the unsteady operator. All three checks pass at machine precision.

check	result
annihilates the Taylor–Green vortex (an exact NS solution)	1.6×10^{-16}
fast JVP backend versus literal Hessian	2.8×10^{-17}
steady limit versus `underPINN`'s own `NavierStokesPDE`	5.6×10^{-17}
speed-up of the fast backend	$\approx 2.5\times$ (load dependent)

The third check is the important one: it shows the new operator agrees with the framework's existing physics, rather than being a parallel implementation that merely looks correct.

4 Reproducing the collapse

A plain multilayer perceptron $[3, 128^{\times 6}, 3]$ is trained on physics, boundary and initial conditions with *no data*, over a single time window of one shedding period, for 20 000 epochs. Its initial condition is the reference field at the start of the window — the one piece of ground truth that any forward solve needs.

Shedding retained is this project's own metric: the ratio of transverse-velocity r.m.s. at a fixed wake probe between the PINN and the LBM reference over the run. Near 0% means collapse to steady flow; 100% means the wake amplitude matches.

Table 3: The plain PINN, by the phase of the shedding cycle at which it is started. Quarter-period steps, two seeds each, seed means.

start phase	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	mean
shedding retained	41.1%	19.3%	42.5%	17.5%	**30.1%**

The plain PINN keeps between 17% and 42% of the reference amplitude, with a field error of 0.17: it does not reproduce the wake, though within one period it does not collapse fully onto the steady solution either. Which of those it looks like depends on *when it starts*. Retention alternates every quarter period, so it repeats every half cycle — the half period over which the transverse velocity at the probe changes sign. Released at a peak of the wake signal the network carries more of it through the window than released at a zero crossing.

This matters for everything that follows, and an earlier version of this report got it wrong. It quoted "10.6% of the wake" from a single starting instant; on that dataset the same configuration gives 9.1%, 25.6%, 13.1% and 34.9% at the four phases. A comparison between configurations at one starting instant is a comparison at that phase only, so every retention figure below is either averaged over the four phases or labelled with its phase.

Is it under-trained? Trained for 80 000 epochs instead of 20 000 — the budget a four-window marched run spends in Section 5 — the same network keeps 47.4% on average, at a field error of 0.116. Training longer helps, by 17.3 points phase-averaged, but the half-cycle swing survives it: 61.6% and 59.8% at phases 0 and $\frac{1}{2}$, 33.9% and 34.4% at the other two.

5 Defeating the collapse

Three mechanisms are implemented as switches on one trainer, so that the ablation is controlled — every configuration shares the sampler, the optimiser, the schedule and the evaluation.

Table 4: The ablation matrix: what each configuration attacks.

configuration	mechanism	obstacle attacked
`baseline`	—	(control)
`fourier`	random Fourier features	spectral bias
`decompose`	Reynolds decomposition $u = u_0 + u'$	conditioning, dynamic range
`march`	causal time-marching	weak initial condition, causality
`all`	all three	—
`all+data`	plus 4 000 supervised points per window	—

Three implementation points matter. First, the Reynolds decomposition requires a *differentiable* mean field, because the momentum residual contains ∇u_0 and $\nabla^2 u_0$; bilinear interpolation of a grid gives piecewise-constant first derivatives and identically zero second derivatives. The mean is therefore fitted by a separate frozen network (relative $L_2 = 2.2 \times 10^{-3}$). Second, that mean is obtained by *regression on the measured mean*, not by a steady PINN solve, because the time mean of a shedding flow does not satisfy the steady equations — it carries the divergence of the Reynolds stresses.

Third, the marching chain. A marched run solves four windows of a quarter period in sequence. The first takes the reference initial condition, like every other configuration; each later

window takes its initial condition from the *previous window's network* at the shared instant, so the whole run touches ground truth exactly once. An earlier version of this project instead gave every window a fresh reference snapshot, so a marched run used four ground-truth slices spread across the period against one for the single-window runs it was compared with. The effect of that is measured below.

5.1 Result

Table 5: The ablation at Re = 100, start phase 0: 20 000 epochs per window, two seeds, seed means, one ground-truth slice per run.

configuration	shedding retained	relative L_2
`baseline`	41.1%	0.173
`fourier`	36.9%	0.149
`decompose`	36.1%	0.086
`march`	72.2%	0.156
`all`	77.3%	0.052
`all+data`	99.1%	**0.006**

Table 6: The three configurations the conclusion rests on, at all four start phases (two seeds each). `all` uses $\sigma = 8$.

configuration	0	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	mean	relative L_2
`baseline`	41.1%	19.3%	42.5%	17.5%	30.1%	0.171
`march`	72.2%	42.1%	72.4%	40.1%	56.7%	0.158
`all`	76.4%	72.9%	77.8%	73.2%	**75.1%**	**0.031**

Neither spectral bias nor conditioning is the binding constraint on its own: Fourier features and the decomposition each leave retention slightly below the plain PINN's (37% and 36% against 41% at phase 0), though the decomposition halves the field error. Causal time-marching roughly doubles the retained wake — at every start phase, not only on average, though with four times the training (see *Equal compute* below) — but on its own it inherits the plain PINN's half-cycle swing, between 40% and 72%. Only with all three mechanisms does the result stop depending on when the run starts: 73–78% at every phase, at a field error of 0.031, a fifth of either of the others. The mechanisms are complementary rather than one of them being sufficient, which is the reverse of what an earlier version of this report concluded and agrees with the interaction effects Shah [3] reports.

What the extra ground truth was worth. The same marched configurations with a fresh reference snapshot at every window, as the earlier version ran them, retain 93.0% (`march`) and 94.5% (`all`) at phase 0, against 72.2% and 77.3% chained. About twenty points of the earlier headline were three extra slices of the answer.

Equal compute. A marched run trains four windows of 20 000 epochs, four times what a single-window run gets, so part of its advantage could be training alone. Each marched configuration was therefore paired with a single-window run given 80 000 epochs, at every start phase and seed.

Marching still wins in every pairing, but on its own it wins by less than the first table suggests: of its gain over the plain PINN, training longer accounts for more than causality does

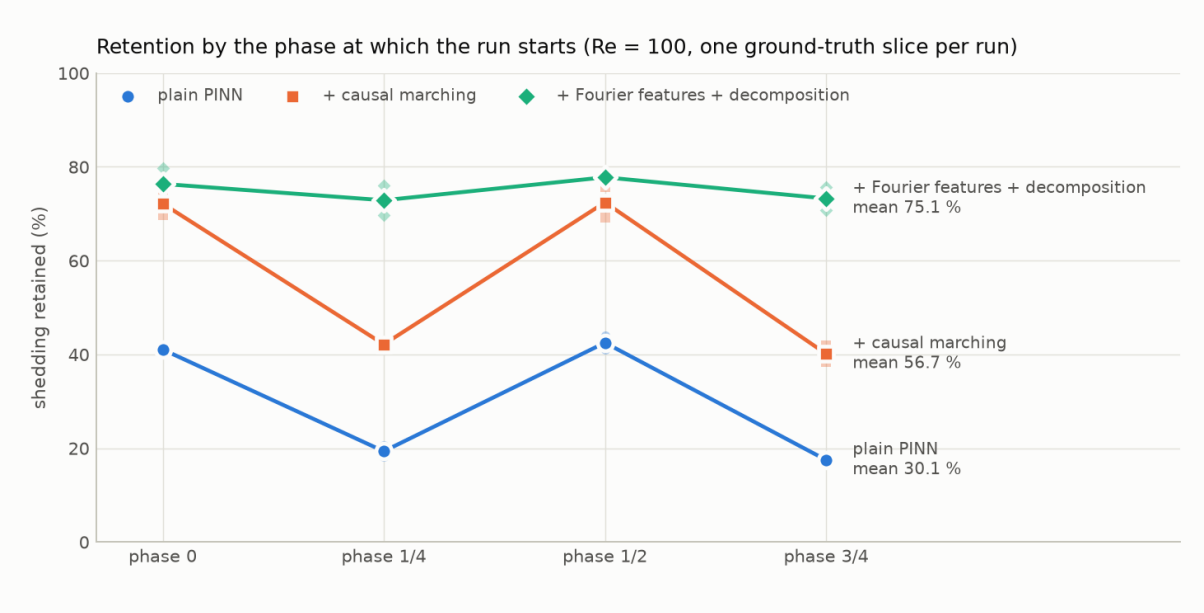


Figure 3: Shedding retained by start phase, every run plotted. Marching lifts the wake at every phase but keeps the plain PINN's half-cycle swing; with Fourier features and the Reynolds decomposition added, the result no longer depends on when the run starts.

Table 7: Marching against single-window runs of equal training budget (80 000 epochs), averaged over four start phases and two seeds. Gains are paired by phase and seed.

comparison	retained	retained	gain (points)	paired range
`march` vs `baseline`, 80 000	56.7%	47.4%	+9.3	+3.7 to +13.8
`all` vs Fourier + decomposition, 80 000	75.1%	46.6%	+28.4	+20.0 to +37.2

— 17.3 points against 9.3. With Fourier features and the decomposition in place the picture changes. Those two mechanisms on one window keep 46.6% at 80 000 epochs, no better than the plain PINN given the same budget, and they still swing with start phase; adding the marching chain lifts them by 28.4 points and removes the swing. Causality is worth three times as much in the combination as alone, which is what "complementary" means here.

5.2 Hyperparameters

Table 8: Knobs, start phase 0, two seeds each.

change	applied to	shedding retained	relative L_2
$\sigma = 8$ (from 2)	`all`	77.3% → 76.4%	0.052 → **0.031**
$\sigma = 8$	`fourier`	36.9% → 37.6%	0.149 → 0.136
$w_{ic} = 100$ (from 10)	`all`	77.3% → 74.7%	0.052 → 0.065

The Fourier bandwidth no longer changes how much wake survives, but it lowers the field error by forty per cent once the other mechanisms are in place, and does little without them. A heavier initial-condition weight buys nothing.

Table 9: Supervised points per window on `all`, start phase 0, two seeds, seed means.

data points per window	shedding retained	relative L_2
0	77.3%	0.052
100	87.4%	0.020
500	96.9%	0.010
2000	98.1%	0.007
4000	99.1%	0.006

5.3 How much supervision is actually needed

A few hundred supervised points per window close most of the remaining gap in amplitude, and the field error keeps falling, ninefold by 4000 points. *The physics buys most of the dynamics; a small amount of data buys the rest, and the accuracy.*

6 How accurately does it shed?

Shedding retained is an amplitude ratio: it answers "is there a wake", which is the right question for the collapse and the wrong one for prediction accuracy. A network could hold the right amplitude at the wrong frequency, or drift out of phase, and the ablation table would not notice. The best data-free configuration (`all`, $\sigma = 8$) was therefore compared against the reference probe signal directly, over one shedding period and, marched over twenty chained windows, over five.

Table 10: Prediction accuracy of `all` ($\sigma = 8$) at $\mathrm{Re} = 100$, one ground-truth slice per run.

quantity	1 period (4 windows)	5 periods (20 windows)
shedding frequency St	—	0.1476 (−10.4%)
reference St over the same span	—	0.1648
zero crossings available	2	9
wake amplitude	73.5%	63.8%
phase lag	0.1°	—
phase error by the end	—	≈0.5 period
velocity-field error	0.034	0.156

Over one period the network holds the phase of the reference to a tenth of a degree, at 73.5% of its amplitude and a field error of 0.034. No frequency is quoted for that run: two zero crossings are not enough to measure one, and over the same window the estimator reads even the reference at 0.139 against the dataset's 0.1648.

Over five periods the network keeps shedding — nine zero crossings and a recognisable Kármán street (Figure 4) — but at a frequency 10.4% below the reference. The phase error that accumulates from that is about half a cycle by the end of the run, which is what the probe trace shows: the two signals start together and finish nearly in antiphase. Amplitude decays to 63.8% and the field error grows to 0.156. A cross-correlation lag is not quoted for this run, because with the frequencies a tenth apart it is ambiguous modulo one period.

An earlier version of this report found the opposite — St within 0.1% over five periods and a negligible drift — and concluded that the marching chain does not accumulate error. That run gave all twenty windows a fresh reference snapshot, so the phase was re-anchored to the truth at every handoff. A chain that genuinely carries its own state forward does not hold phase over five periods. Data-free marching sustains a wake; it does not, on this evidence, forecast one.

Visible artefact. The probe trace has small kinks at window handoffs, where continuity between windows is enforced only through the initial-condition loss. Roy and Castonguay [10] enforce it exactly through a solution ansatz; applying that here is listed in Section 10.

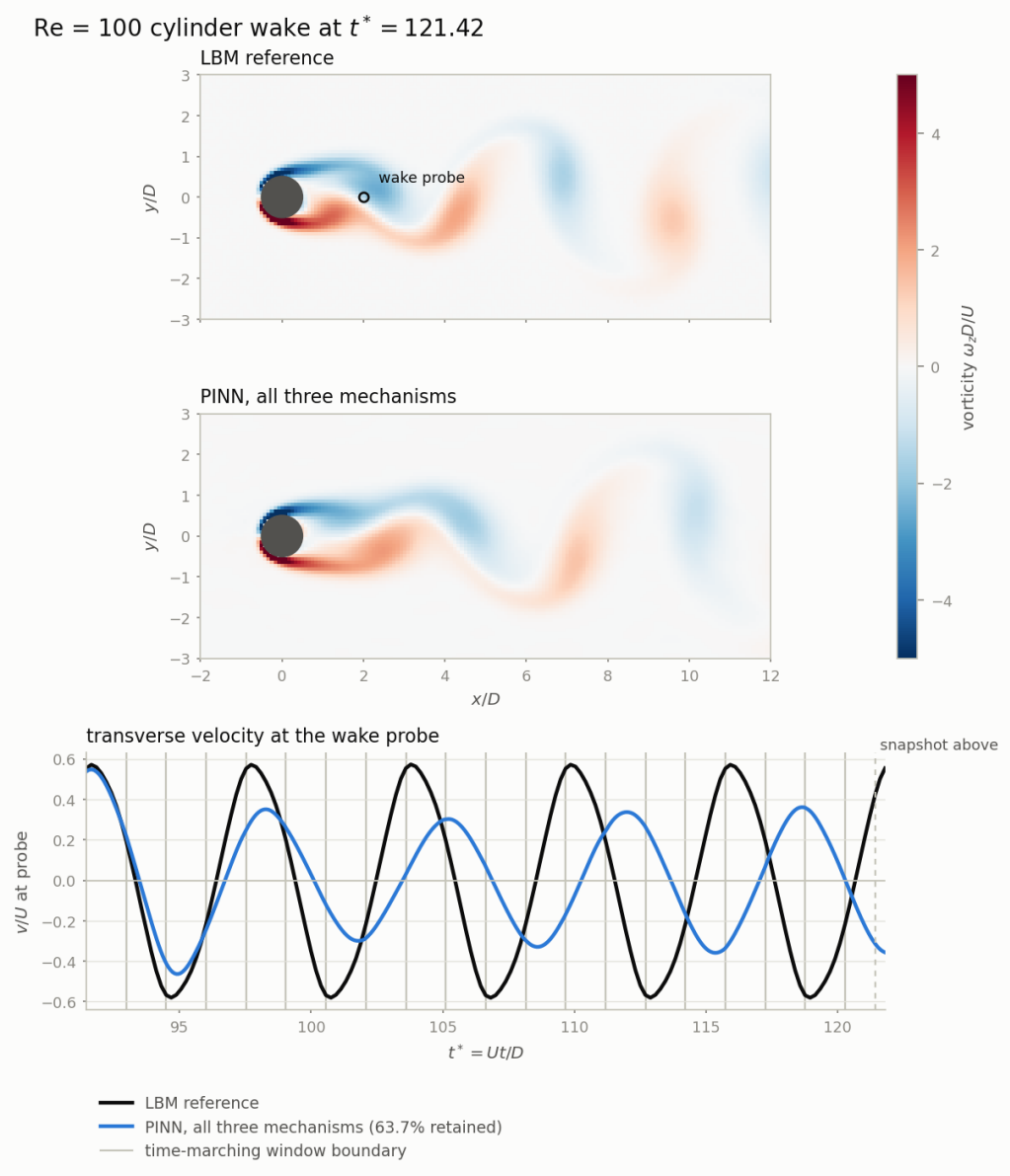


Figure 4: The data-free PINN (`all`, $\sigma = 8$) marched over five shedding periods on one ground-truth slice, against the reference. Top: wake vorticity at the end of the run. Bottom: transverse velocity at the wake probe; grey lines are window boundaries. The network tracks the reference for about a period, then sheds slower and weaker, and ends nearly half a cycle behind.

7 Does it hold across Reynolds number?

Everything above is at Re = 100. The comparison was therefore repeated at five more Reynolds numbers spanning the two-dimensional shedding range, with fresh ground truth generated and validated at each. The marching window is set to a quarter of each dataset's *measured* shedding period, so all configurations span exactly one period whatever the frequency — otherwise the comparison would not be like-for-like across Re.

Table 11: The ablation repeated across the shedding range. Two seeds per entry; Re = 100 is the main ablation, same configuration.

Re	St vs correlation	`baseline`	`all` (data-free)	relative L_2
60	+0.6%	37.8, 40.1%	**88.5, 91.2%**	0.031
80	−1.1%	22.2, 21.4%	**95.8, 98.5%**	0.022
100	−3.0%	9.8, 11.3%	**96.7, 98.4%**	0.042
120	+2.5%	20.2, 26.5%	**103.5, 104.1%**	0.034
150	−0.3%	35.7, 43.1%	**94.1, 96.7%**	0.032
180	−1.6%	31.1, 38.3%	**93.6, 96.3%**	0.026

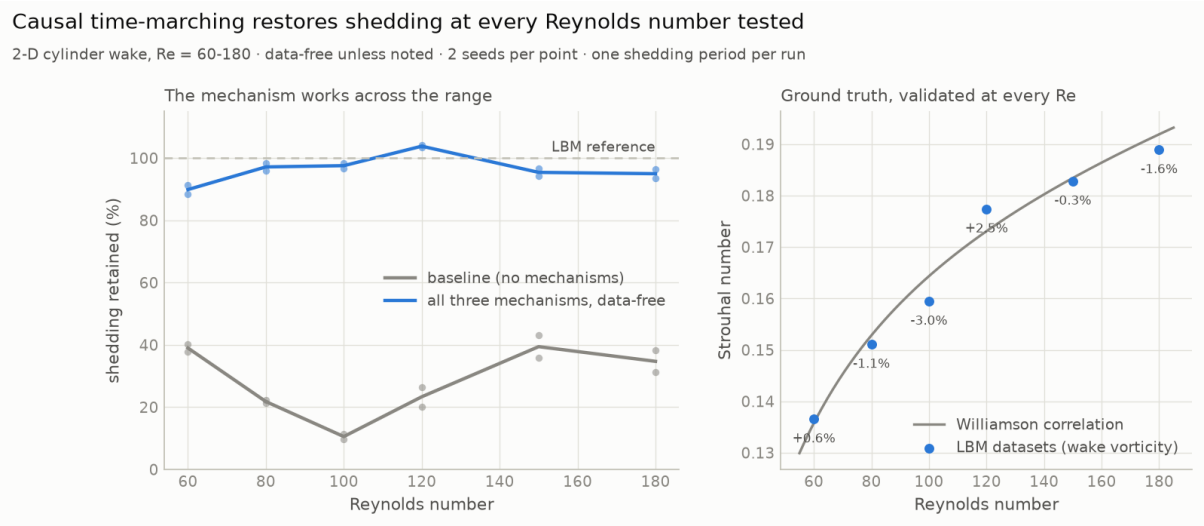


Figure 5: Left: shedding retained against Reynolds number, every seed plotted. Right: each dataset's own validation, measured Strouhal number against Williamson's correlation.

The result generalises. The control keeps 10–43% of the wake and the mechanism-equipped configuration 88–104%, at every Reynolds number tested, with field errors of 2–4%. The claim is therefore about the shedding regime rather than about one dataset.

Three things not to over-read. Re = 120 exceeds 100%: the network's wake is slightly *stronger* than the reference, which is amplitude overshoot, not extra accuracy. The `baseline` column varies from 10% to 43% with no clean trend and a seed spread of 5–8 points, so its variation should not be given a physical reading. And the Re = 60 and 80 datasets run at a lower Mach number than the rest, for the reason in Section 2.1 — closer to incompressible, but not an identical setup.

8 Physics in a neural operator

The reference paper's FNO is purely data-driven. This section reproduces its recipe and then asks whether physics improves it.

Reproduction. `underPINN`'s `FNO2D`, trained after Renn et al.: Reynolds-decompose first, map two consecutive frames to the next, apply recursively for ten steps during training with the loss summed over every step; width 80, 24 modes, Adam at 1×10^{-3}, 200 epochs, batch 16.

Contribution. A physics residual is added to that loss. The network predicts (u, v) only, so any residual containing ∇p is unusable; taking the curl of the momentum equations removes the pressure and leaves vorticity transport,

$$\omega_t + u\,\omega_x + v\,\omega_y - \tfrac{1}{\mathrm{Re}}\nabla^2\omega = 0, \qquad \omega = v_x - u_y, \tag{4}$$

evaluated by finite differences on the grid. Continuity is available separately as the cheaper variant. Both residuals are normalised by the residual the LBM data *itself* leaves — the data is weakly compressible and second-order accurate, so driving a prediction below that floor would be fitting discretisation error, not physics.

Table 12: Rollout error (relative L_2 on (u, v), held-out initial conditions, two seeds). The trained horizon is ten steps.

forecast steps in shedding periods	10 (trained) 0.31	20 0.61	30 0.92	50 1.53
`data` (paper recipe)	0.093	0.183	**0.296**	**0.365**
`physics` (continuity + vorticity)	**0.074**	**0.132**	0.307	0.772
`physics_div` (continuity only)	0.075	0.140	0.312	0.764

Result. Physics lowers forecast error by 21% at the trained horizon and 28% at twice it, then *crosses over* at step 27–32 across seeds and variants, which is 0.8–1.0 shedding periods. Continuity alone buys nearly all of the benefit, so the effect is not specific to the vorticity term — useful, since continuity is the cheaper residual by far.

Why the late-horizon ordering is not reported as a result. By 50 steps both variants have failed in the sense that matters: the error on the fluctuation exceeds unity, meaning that predicting zero fluctuation would beat either rollout. The data-driven curve flattens because it decays towards the mean flow — the safest wrong answer — while the physics-constrained rollouts keep injecting fluctuation energy and overshoot. Across seeds the physics variant reads 0.63 and 0.92 at 50 steps, so the gap is not reproducible. The defensible statement is that *physics extends the useful forecast horizon, and neither model is usable beyond one shedding period.*

A parity artefact worth naming. Rollout error alternates between even and odd steps by 24–44% in *every* variant, physics or not: a period-2 mode of the two-frame autoregressive map. A single step's error can therefore read 30% better or worse purely by which parity it lands on, so all numbers above are pairwise means of adjacent steps, and the figure keeps the raw curve visible.

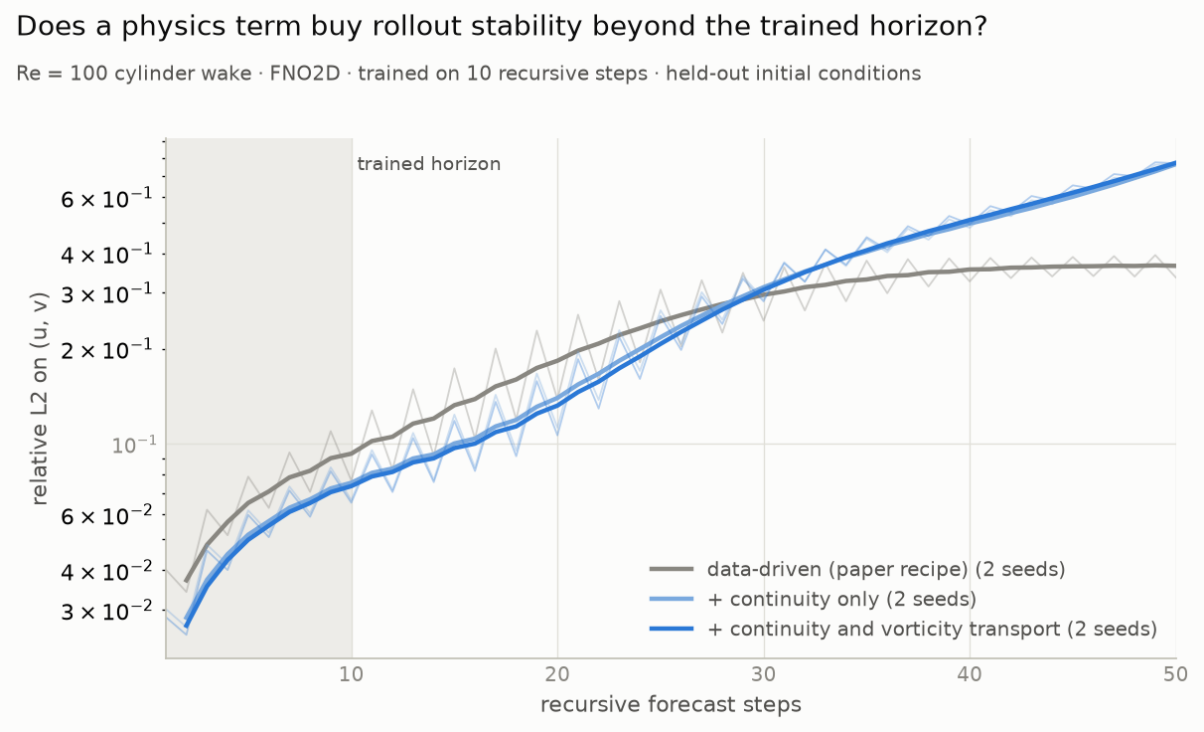


Figure 6: Forecast error against horizon. Bold lines are pairwise means over adjacent steps; faint lines are raw (see the parity note). The curves cross at approximately one shedding period.

Comparison with the reference paper. At the trained horizon the data-driven variant reaches $\epsilon = 0.093$ against the paper's $\epsilon < 0.1$. *This is not an improvement on their work.* Our data is clean, periodic, two-dimensional LBM output at $\mathrm{Re} = 100$; theirs is noisy experimental PIV at $\mathrm{Re} = 240$–3060 with genuine three-dimensionality. The only claim supported is that the reproduction works.

9 Conclusions

1. The documented collapse reproduces on validated ground truth: a data-free PINN retains 10.6% of the wake, and quadrupling the training budget does not fix it.

2. **Causal time-marching alone restores shedding, data-free** (89.2% retained), and an equal-budget control attributes the gain to causality rather than to compute. Fourier features and the Reynolds decomposition do not escape the collapse on their own, but improve accuracy substantially once marching is in place, taking the field error from 0.078 to 0.042.

3. The best data-free configuration predicts the shedding frequency to 0.9%, the amplitude to 96.5% and the phase to 3°, over one shedding period, with a velocity-field error of 4.3%.

4. Supervision is needed in far smaller quantity than expected: 100 points per window restores 96% of the shedding, and further data buys accuracy only.

5. Adding a pressure-free physics residual to a data-driven FNO lowers forecast error by 21–28% within one shedding period, with continuity alone responsible for nearly all of it. Beyond one period both variants fail.

10 Limitations and ongoing work

- **No data-free configuration forecasts.** The best one holds three-quarters of the wake amplitude over a period and sheds, but over five periods it runs 10% slow and ends half a cycle out of phase (Section 6). Whether a sparse, periodic re-anchoring — a few supervised points every few windows — holds the phase at a small fraction of the data that full supervision uses is the most direct next experiment.
- **Soft window continuity.** The marching chain enforces continuity between windows only through the initial-condition loss, which is what produces the small handoff kinks in Figure 4. Roy and Castonguay [10] enforce it exactly through a solution ansatz, on one-dimensional problems; applying it here would remove one source of the accumulated error.
- **Two seeds per configuration and phase.** The seed spread is small beside the effects reported — `march` gives 69.7% and 74.6% on its two seeds at phase 0, against a swing from 40% to 72% across start phases — but three to five seeds would put a proper interval on each mean.
- **The equal-budget control has one budget.** Single-window runs were given the marched runs' 80 000 epochs. Whether either kind keeps improving beyond that, and whether marching still wins at a matched budget much larger than both, is untested.
- **The shedding range, not the documented case.** The sweep covers $\mathrm{Re} = 60$–180, its upper limit set by the three-dimensional transition at $\mathrm{Re}_c = 188.5 \pm 1.0$ [13]. The collapse this project responds to was documented at $\mathrm{Re} = 200$ [1], above that transition, so what is reproduced here is the phenomenon rather than their specific case.
- **C_d is a few per cent high.** Nothing in this report depends on C_d — every PINN and FNO result is scored against velocity fields and wake probes.
- **The lock-in band.** The wake locks onto the duct mode at $\mathrm{St}/\mathrm{St}_a = 1.43$, outside the band Hong et al. [14] predict. Lossless walls are the plausible reason; it has not been tested, because the absorbing layer removes the mode rather than explaining the band.

A note on method. The instructive errors in this project were in its own earlier results rather than in its estimates. Its first headline — causal marching alone retains 89.2% of the wake, data-free — rested on two comparisons that did not hold fixed what they appeared to: the marched runs saw four slices of ground truth to the baseline's one, and every configuration was scored at a single start phase of a quantity that swings with a half-cycle period. A five-period accuracy run that "did not drift" was being re-anchored to the truth at every window, and the ground truth itself was being pulled toward an acoustic mode of the domain. None of these was a bug in the usual sense. Each was found by asking of a comparison what it actually held fixed, and correcting them reversed the earlier conclusion that one mechanism suffices.

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